

Notes: Benson, Li, and Shue (AER, 2026)

“Potential” and the Gender Promotion Gap

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1 Big picture

- **Question.** Why are women promoted less often than men even when they perform at least as well? The paper studies whether subjective assessments of worker *potential*, rather than realized performance, are an important source of the gender promotion gap.
- **Institutional setting.** The paper uses detailed personnel data from a large North American retail chain that evaluates salaried management-track workers with a “Nine Box” system. Supervisors rate each worker on two dimensions: current *performance* and future *potential*. Each dimension takes values 1, 2, or 3.
- **Why this setting is powerful.** Most firms informally distinguish between “what the worker has done” and “what the worker could become.” The Nine Box system makes that distinction observable. It lets the authors separate backward-looking assessments of performance from forward-looking assessments of potential.
- **Main descriptive fact.** Women receive *higher performance ratings* than men but *lower potential ratings*. So the promotion gap is not coming from women doing worse in their current jobs.
- **Main promotion result.** Women are about 13% less likely to be promoted than men. Differences in potential ratings explain about half of this promotion gap, depending on the specification.
- **Main bias result.** Lower potential ratings for women do *not* accurately forecast lower future contributions. Conditional on current performance and potential ratings, women receive higher future performance ratings than comparable men. This is true both in the full sample and among workers who are actually promoted.
- **Persistence of bias.** Even after women outperform men with the same initial ratings, managers continue to assign women lower future potential ratings. The firm does not seem to update beliefs about women’s potential in response to realized success.
- **Mechanism 1: strategic retention.** Men are more likely to leave the firm when passed over for promotion. Managers appear to use potential ratings partly to retain likely quitters. Because high quit risk is concentrated more among men, this channel mechanically favors men.
- **Mechanism 2: stereotyping.** The gender potential gap is larger in counties with fewer women in management, larger gender wage gaps, and lower female educational attainment. This pattern is consistent with role-congruity or stereotyping channels.
- **What the paper does *not* claim.** The paper is not estimating the causal effect of implementing Nine Box. Instead, Nine Box data are used as a window into how firms think about advancement and “high potential” status more broadly.
- **Misallocation result.** Using an instrument for promotion opportunities, the authors show that *marginally promoted women* outperform marginally promoted men in subsequent performance ratings. This is a Becker-style outcomes test suggesting the firm promotes too few women at the margin.
- **Policy takeaway.** Simply eliminating potential ratings would reduce gender disparities but would also throw away useful information, because potential ratings do forecast future performance on average. The challenge is therefore to *debias* assessments of potential rather than discard them altogether.

- **Bottom line.** The paper argues that a meaningful part of the gender promotion gap comes from subjective, biased forecasts of who looks like leadership material, not from inferior female performance.

2 Data and measurement (key objects)

2.1 The firm's evaluation system: the Nine Box

The core institutional object is the firm's **Nine Box** grid. Every management-track worker is rated along two dimensions:

- **Performance:** how well the worker is doing in the current role.
- **Potential:** the worker's expected future contribution, either through stronger performance in the current role or through leadership in a higher-level role.

Each dimension takes values 1 (low), 2 (medium), or 3 (high), producing a 3-by-3 grid. The paper emphasizes that performance is intentionally backward-looking and more verifiable, while potential is inherently forward-looking and therefore much more subjective.

The box corresponding to low performance and high potential is reserved for new hires. Because that cell is mechanically assigned, the authors drop those observations from the main analysis.

2.2 Sample coverage

The data cover **29,809** full-time, salaried, management-track workers employed between **2011 and 2015**. The main panel is at the worker-month level and contains **900,209 worker-month observations**.

Key coverage facts:

- more than **4,000 locations**,
- **49 business units**,
- and **8,964 promotion events**.

Workers come from both corporate headquarters and the firm's retail locations. The sample excludes entry-level workers such as cashiers because these workers are not part of the management-track population evaluated with Nine Box.

2.3 How promotions are measured

The paper constructs promotions using standardized job titles and salary changes. A worker is classified as promoted if, in the following month, either:

1. the worker moves to a job title associated with higher average compensation, or
2. the worker changes title and receives a personal salary increase exceeding 5%.

The main promotion outcome is converted to an **annualized percent** by multiplying the monthly

promotion indicator by 1,200. This way, coefficients can be read directly as percentage-point changes in annual promotion rates.

2.4 Key variables

Promotion. Indicator for whether the worker is promoted in the next month, annualized.

Salary. Annual salary measured in dollars, often entered in logs in regressions.

Performance rating. Current Nine Box performance rating taking values 1, 2, or 3.

Potential rating. Current Nine Box potential rating taking values 1, 2, or 3.

Future performance and future potential. Ratings measured 12 months ahead. These are used to assess whether current potential ratings accurately forecast later outcomes.

Leaves of absence. Number of months of leave taken in the past or future. These are used to test whether lower female potential ratings reflect anticipated interruptions.

Risk of loss. A firm-assigned 3-point measure of expected attrition risk, available in a three-year subsample. This variable helps test whether managers reward workers who are likely to quit.

Attrition. Indicator for whether the worker leaves the firm in the next month, annualized.

2.5 Summary statistics

The main descriptive moments are:

- 41.2% of workers are female,
- the mean annualized promotion rate is 11.9%,
- mean salary is about \$70,691,
- mean performance rating is 2.18,
- mean potential rating is 1.43,
- mean age is 44.4 years,
- and mean tenure is 171 months.

The raw correlations already preview the main findings: being female is negatively correlated with promotions, salary, and potential ratings, but positively correlated with performance ratings.

2.6 Female representation in the hierarchy

The firm exhibits a classic glass-ceiling pattern.

- Women are 56% of entry-level field workers.
- They are 48% of department managers.
- They are 35% of store managers.
- They are only 14% of district managers.

Looking across the entire salaried workforce with Nine Box ratings, women are about 51% of the bottom pay decile but only 29% of the top pay decile.

Interpretation. Female underrepresentation is not coming from the entry point alone. It widens as one moves up the hierarchy.

2.7 What the raw rating distributions show

Among male workers, performance ratings cluster in the middle: about 70% receive a 2, 23% receive a 3, and 7% receive a 1. Potential ratings are much harsher: about 60% of men receive the lowest potential rating, 35% receive the middle rating, and only 5% receive the highest rating.

Relative to men:

- women are 7.4% more likely to receive the top performance rating,
- women are 21% less likely to receive the lowest performance rating,
- women are 12% more likely to receive the lowest potential rating,
- women are 15% less likely to receive the middle potential rating,
- and women are 28% less likely to receive the top potential rating.

The paper also highlights that women are 33% more likely than men to fall in the “high performance, low potential” box, sometimes informally called the *workhorse* category.

Interpretation. The firm is not saying women underperform. It is saying women look less like future leaders.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline promotion regression

The paper’s starting point is a linear probability model for promotion:

$$\text{Promotion}_{it} = a_1 \text{Female}_i + a_2 X_{it} + \delta_y + \varepsilon_{it}. \quad (1)$$

Here:

- Promotion_{it} is an indicator for whether worker i is promoted in the next month, annualized to percent,
- Female_i is an indicator for gender,
- X_{it} includes controls such as performance ratings, demographic variables, location fixed effects, and business-unit fixed effects,
- and δ_y are fiscal-year fixed effects.

Interpretation of a_1 . Without controls, a_1 is the overall gender gap in promotions. With controls, a_1 is the remaining gender gap after conditioning on observed worker characteristics and workplace assignment.

3.2 Potential and performance rating regressions

To document that women are treated differently on the two dimensions of Nine Box, the paper estimates regressions of the form:

$$\text{Potential}_{it} = b_1 \text{Female}_i + b_2 X_{it} + \delta_y + u_{it}, \quad (2)$$

$$\text{Performance}_{it} = c_1 \text{Female}_i + c_2 X_{it} + \delta_y + v_{it}. \quad (3)$$

Interpretation.

- A negative b_1 means women receive lower potential ratings.
- A positive c_1 means women receive higher performance ratings.

The key empirical fact is that both happen at the same time.

3.3 How much of the promotion gap is explained by potential?

To see whether potential is the channel linking gender to promotion, the paper augments the promotion regression with current potential ratings:

$$\text{Promotion}_{it} = \pi_1 \text{Female}_i + \pi_2 \text{Potential}_{it} + \pi_3 \text{Performance}_{it} + \pi_4 X_{it} + \delta_y + e_{it}. \quad (4)$$

Interpretation.

- If adding potential sharply shrinks the female coefficient, then differences in potential ratings explain a sizable portion of the gender promotion gap.
- If the coefficient on potential is larger than the coefficient on performance, then the firm uses forward-looking judgments more strongly than backward-looking performance when deciding whom to promote.

That is exactly what the paper finds.

3.4 Testing whether lower female potential is justified

The paper next asks whether women receive lower potential ratings because they truly have lower future contributions. The main test regresses future performance on current ratings and gender:

$$\text{Perf}_{i,t+12} = \theta_1 \text{Female}_i + \theta_2 \text{Potential}_{it} + \theta_3 \text{Performance}_{it} + \theta_4 X_{it} + \eta_{it}. \quad (5)$$

This regression is run both in the full sample and in the subsample of workers who are promoted.

Interpretation of θ_1 .

- If $\theta_1 = 0$, then women and men with the same current ratings have the same future performance.
- If $\theta_1 > 0$, then women outperform men with the same current ratings, which means women's potential is being understated.

The paper finds $\theta_1 > 0$.

3.5 Testing whether managers update beliefs

To study persistence, the authors run a parallel regression with future potential as the dependent variable:

$$\text{Potential}_{i,t+12} = \phi_1 \text{Female}_i + \phi_2 \text{Potential}_{it} + \phi_3 \text{Performance}_{it} + \phi_4 X_{it} + \xi_{it}. \quad (6)$$

Interpretation of ϕ_1 . If women continue to receive lower future potential ratings even after outperforming men with the same initial ratings, then the firm is not updating beliefs in an unbiased way.

That is the core result of Table 6, panel B.

3.6 Leaves and family-related interruptions

To test whether lower female potential reflects anticipated interruptions, the paper estimates specifications that relate potential ratings to past and future leave-taking:

$$\text{Potential}_{it} = \rho_1 \text{Female}_i + \rho_2 \text{PastLeave}_i + \rho_3 \text{FutureLeave}_i + \rho_4 (\text{Female}_i \times \text{Leave}) + \rho_5 X_{it} + \delta_y + \omega_{it}. \quad (7)$$

Interpretation. If women are penalized because managers expect family-related absences, then controlling for leave-taking should explain much of the female potential gap, especially among younger workers.

The paper finds only limited support for this channel.

3.7 Attrition, risk of loss, and strategic retention

A second mechanism is that managers may use potential ratings to retain workers who threaten to quit. The paper studies regressions where the outcomes are:

- actual future attrition,
- firm-assigned risk-of-loss ratings,
- future potential ratings,
- next-month promotions,
- future salary,
- and future performance.

The key right-hand-side variable is the firm's risk-of-loss measure, alongside gender and current

ratings.

Interpretation. If managers strategically retain likely quitters, then workers assessed as higher risk-of-loss should receive higher potential ratings, more promotions, and higher future pay even if they do not deliver higher future performance.

This is precisely the pattern the paper documents.

3.8 Promotion desirability and worker aspirations

To examine whether women simply want fewer promotions, the paper re-estimates promotion regressions for three especially desirable categories:

- promotions that remain in the same metropolitan area,
- promotions with salary raises greater than 25%,
- and promotions that do not add supervisory responsibilities.

Interpretation. If lower female ambition were the main driver, gender gaps should shrink sharply in these especially appealing promotion categories.

The paper finds the opposite: sizable gaps remain, and potential ratings still explain a large share of them.

3.9 Stereotyping and local gender norms

The paper links worker locations to county-level gender inequality measures and estimates interactions such as:

$$\text{Potential}_{it} = \kappa_1 \text{Female}_i + \kappa_2 \text{CountyIneq}_c + \kappa_3 (\text{Female}_i \times \text{CountyIneq}_c) + \kappa_4 X_{it} + \delta_y + \zeta_{it}. \quad (8)$$

The county variables include:

- the share of managers who are men,
- the county gender wage gap,
- and female educational attainment.

Interpretation of κ_3 . If the female potential gap is bigger in places with less gender equality, that is consistent with a stereotyping channel rather than a pure productivity channel.

3.10 Becker outcomes test using an IV for marginal promotions

To test whether the firm underpromotes women, the paper uses an instrument for promotion opportunities. Let Z_{it} denote the average promotion rate for workers with the same job title in the same year, leaving out workers in worker i 's own office or store.

The core IV-style equations are estimated separately by gender:

$$Y_{it} \times P_{it} = \alpha_0 + \alpha_1 P_{it} + X'_{it} \alpha + \varepsilon_{it} \quad \text{if female,} \quad (9)$$

$$Y_{it} \times P_{it} = \beta_0 + \beta_1 P_{it} + X'_{it} \beta + \nu_{it} \quad \text{if male.} \quad (10)$$

Here:

- P_{it} is promotion,
- Y_{it} is a future rating outcome, either future performance or future potential,
- and P_{it} is instrumented by Z_{it} .

Interpretation. The IV coefficients identify the future outcomes of workers who are promoted on the margin when promotion opportunities are more plentiful. Comparing female and male compliers provides a Becker-style outcomes test for discrimination.

3.11 Counterfactual promotion policy

Finally, the paper simulates a policy that ignores *gender* and *potential ratings* when assigning promotions. The fitted baseline promotion rule is estimated from current data; then the coefficients on gender and potential are set to zero and promotion probabilities are rescaled so that the aggregate promotion rate stays fixed.

Interpretation. This exercise quantifies the trade-off between:

- **equity**: reducing gender gaps in promotion,
- and **efficiency**: preserving the average future performance of promoted workers.

4 Identification and instruments (what makes the estimates credible)

4.1 The main descriptive design is transparent about what it identifies

The paper is explicit that the early regressions are not causal estimates of the effect of using Nine Box. Instead, they document how the firm’s own subjective promotion technology works. That matters because the question is not “does Nine Box cause bias?” but rather “does observed managerial assessment of potential contribute to gender gaps?”

4.2 Why the potential-bias claim is stronger than a raw gender gap

A raw promotion gap by itself could reflect omitted productivity differences. The paper’s stronger claim comes from a sequence of comparisons:

- women get higher current performance ratings,
- women get lower current potential ratings,
- women later outperform men conditional on the same current ratings,

- yet women still receive lower future potential ratings.

That sequence is hard to reconcile with the idea that managers are simply making accurate forecasts.

4.3 The future-performance test is the key credibility exercise

Table 6 is central. Potential ratings do predict future performance on average, which means the variable is not pure noise. But conditional on current ratings, women have *higher* future performance than men. This is the most direct evidence that women's potential is being undervalued rather than accurately forecast.

This is especially compelling because the same result holds:

- in the full sample,
- and in the sample of promoted workers, where the future rating reflects performance in a new role.

4.4 Why the leave-of-absence explanation is weak

If managers were rationally discounting women because they expected more career interruptions, then controlling for leave-taking should largely eliminate the female potential gap. The paper shows that it does not.

In fact, the leave penalty is often stronger for men than for women. That suggests the leave channel is at most a small part of the story and is itself shaped by gender norms.

4.5 Why the attrition story is not a simple productivity story

Women are less likely to leave the firm, and managers appear to know this because women receive lower risk-of-loss ratings. But higher risk-of-loss workers receive better future treatment in potential, promotions, and pay without delivering higher future performance.

This is exactly what one would expect if the firm uses ratings strategically to retain workers who threaten to exit, not if higher risk of loss proxies for better underlying talent.

4.6 Why the stereotyping interpretation is plausible

The paper shows that the female potential gap is larger in counties with:

- fewer women in management,
- larger gender wage gaps,
- and lower female educational attainment.

At the same time, analogous interactions for current performance ratings are much weaker or insignificant. That difference matters. It suggests the county patterns are about beliefs regarding future leadership suitability rather than local differences in women's actual on-the-job performance.

4.7 What identifies the IV outcomes test

The IV analysis uses variation in promotion opportunities across job-title-by-year cells, excluding the worker's own office or store. The idea is that some workers face temporarily greater promotion availability for reasons unrelated to their own quality.

Why this is credible:

- the instrument is leave-one-location-out, so an individual's own promotion does not mechanically raise the instrument,
- the instrument varies at the job-title-by-year level, which captures expansionary opportunities rather than a single manager's view of one worker,
- future performance is residualized by job-title-by-year fixed effects, so the comparison is within the same promoted role and year,
- and the final object of interest is the *difference* between female and male compliers, which is less sensitive to common shocks than the levels themselves.

4.8 Remaining limitations

The paper is careful about several limitations.

- It studies one firm, albeit a very large one.
- Performance ratings are more objective than potential ratings, but they are still not perfectly objective.
- County-level and manager-level heterogeneity results are suggestive because worker-manager matches are not random.
- The counterfactual policy exercise is partial equilibrium and does not capture all retention, hiring, or search costs.
- The IV exercise identifies marginal promoted workers, not the effect of promoting every additional woman.

4.9 Relation to the literature

The paper sits at the intersection of several literatures:

- the gender promotion and glass-ceiling literature,
- research on subjective performance evaluation and discrimination,
- the literature on the Peter Principle and promotion mistakes,
- and work on talent hoarding, retention, and internal labor markets.

Its distinctive contribution is to show that *subjective forecasts of potential* are a quantitatively important and empirically observable source of gender gaps in advancement.

5 Tables and figures

5.1 Figure 1 and Table 1: the Nine Box system and the basic sample

		Performance		
		1 (Low)	2 (Medium)	3 (High)
Potential	3 (High)	New hire	Delivering, strong potential	High-performing, top talent
	2 (Medium)	Potential mismatch	Delivering, promotable	High-performing, promotable
	1 (Low)	Underperforming	Delivering	High-performing, critical resource

FIGURE 1. NINE BOX RATINGS AND LABELS

Notes: The table reports facsimiles of the labels used by our data provider. The box for low performance, high potential is explicitly used for new hires.

TABLE 1—SUMMARY STATISTICS AND CORRELATIONS						
<i>Panel A. Data coverage</i>						
Workers	29,809	Locations	> 4,000			
Months (2011–2015)	58	Business units	49			
Worker-months	900,209	Promotions	8,964			
	Mean	SD	p25	p50	p75	
<i>Panel B. Summary statistics</i>						
(a) Female	0.412	0.492	0	0	1	
(b) Promotion (annualized percent)	11.9	119.148	0	0	0	
(c) Salary (annual dollars)	70,691	101,974	45,000	59,188	85,000	
(d) Performance rating	2.18	0.536	2	2	2	
(e) Potential rating	1.429	0.578	1	1	2	
(f) Age	44.4	10.834	35.8	45	53	
(g) Tenure (months)	171.2	139.534	53	138	253	
	(a)	(c)	(d)	(e)	(f)	
<i>Panel C. Correlations</i>						
(a) Female	1					
(b) Promotion (annualized percent)	−0.007	1				
(c) Salary (annual dollars)*	−0.132	−0.019	1			
(d) Performance rating	0.032	0.025	0.206	1		
(e) Potential rating	−0.071	0.048	0.176	0.088	1	
(f) Age*	0.037	−0.052	0.193	0.015	−0.271	1
(g) Tenure (months)*	0.072	−0.039	−0.021	0.097	−0.215	0.465

Note: Asterisks denote that salary, age, and tenure are computed as log variables in panel C and subsequent analyses.

Read:

- Figure 1 lays out the 3-by-3 Nine Box grid and the labels attached to different cells.
- Table 1 shows 29,809 workers, 900,209 worker-months, more than 4,000 locations, and 8,964 promotions.
- The average annualized promotion rate is 11.9%, the mean performance rating is 2.18, and the mean potential rating is only 1.43.
- Women are 41.2% of the sample and are negatively correlated with salary, promotions, and potential, but positively correlated with performance.

Economic message. The firm already treats “potential” as a scarce category. The low average potential rating means that getting labeled high potential is unusual and therefore potentially very important for advancement.

5.2 Figure 2 and Tables 2–3: hierarchy, promotion, and pay

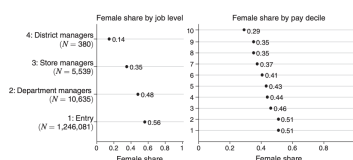


FIGURE 2. FEMALE SHARES IN THE ORGANIZATIONAL HIERARCHY

Notes: The left panel reports the female share among retail operations workers. Department managers include all managers junior to the location's head manager, including associate managers overseeing departments and salaried assistant general managers. Counts include the number of unique workers who held a job at each level within our sample period. The right panel reports the female share among all workers who receive Nine Box ratings. This population includes all regular, salaried workers, including corporate workers and field workers at the level of department managers and above, and excludes entry-level retail workers. The deciles are sorted by regular annual salaries.

TABLE 2—GENDER GAP IN PROMOTIONS					
Promoted	(1)	(2)	(3)	(4)	(5)
1: Female	−1.644 (0.267)	−1.337 (0.266)	−1.027 (0.255)	−1.079 (0.280)	−0.981 (0.282)
Performance rating					
2 = Med		6.498 (0.325)	6.061 (0.331)	5.417 (0.378)	5.369 (0.378)
3 = High		11.35 (0.454)	11.74 (0.426)	10.99 (0.482)	10.90 (0.482)
Fiscal year FEs	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes
Location FEs	Yes	Yes	Yes	Yes	Yes
Business unit FEs	Yes	Yes	Yes	Yes	Yes
Observations	900,209	900,209	900,209	900,209	900,209

Note: This table reports a linear probability model for promotions. The dependent variable takes a value of 1/200 if the worker is promoted in the following month, and 0 otherwise so that coefficients represent annualized percent. The dependent variable mean is 11.9 percent. The omitted category for the performance rating is 1 = Low. Demographic controls include log age, log tenure, and race fixed effects (white, Black, Asian, Hispanic, and other). Standard errors are clustered by worker.

TABLE 3—GENDER PAY GAP AND THE ROLE OF PROMOTIONS				
log salary	(1)	(2)	(3)	(4)
Female	−0.118 (0.00566)	−0.0364 (0.00391)	−0.159 (0.00409)	−0.0714 (0.00285)
Potential rating				
2 = Med			0.0760 (0.00462)	0.0412 (0.00215)
3 = High			0.181 (0.00743)	0.0906 (0.00508)
Performance rating				
2 = Med			0.0427 (0.00437)	0.0295 (0.00296)
3 = High			0.149 (0.00539)	0.0881 (0.00369)
Fiscal year FEs	Yes	Yes	Yes	Yes
Job level × year FEs	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
Location FEs	Yes	Yes	Yes	Yes
Business unit FEs	Yes	Yes	Yes	Yes
Observations	899,726	899,726	899,726	899,726

Note: This table reports regressions of log salary on the female indicator, performance rating indicators (the omitted category is 1 = Low), potential rating indicators (the omitted category is 1 = Low), fiscal year fixed effects, and/or job level interacted with fiscal year fixed effects. Data include 87 FLSA-exempt field sales managers working only on commission. Standard errors are clustered by worker.

Read:

- Figure 2 shows female representation falling sharply as job level and pay rise.

- Table 2 shows women are about 1.64 percentage points less likely to be promoted annually in the raw data, or about 13% less likely relative to men's rate.
- Controlling for current performance actually makes the gender promotion gap larger because women receive higher performance ratings.
- Table 3 shows an overall salary gap of roughly 12.5%, and about 70% of that gap is explained by differences in job level.

Economic message. Most of the pay gap in this setting works through job assignment and promotion, not simply through lower pay within the same role.

5.3 Figure 3 and Tables 4–5: the divergence between performance and potential

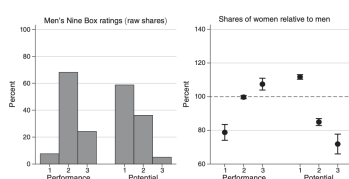


FIGURE 3. GENDER GAP IN NINE BOX RATINGS
Notes: The left panel represents the distribution of Nine Box performance ratings and potential ratings assigned to male workers. The right panel represents the share of women relative to men who receive the rating in the horizontal axis. Vertical brackets represent 95 percent confidence intervals.

	Potential rating		Performance rating	
	(1)	(2)	(3)	(4)
Female	-0.083 (0.097)	-0.0069 (0.082)	0.0343 (0.083)	0.0233 (0.085)
Fiscal year FEs	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
Location FEs	Yes	Yes	Yes	Yes
Business unit FEs	Yes	Yes	Yes	Yes
Observations	900,209	900,209	900,209	900,209
FV mean	1.438	1.428	2.1799	2.1799

Notes: This table reports OLS regressions of Nine Box potential and performance ratings (1, 2, or 3) on the female indicator. Standard errors are clustered by worker.

	(1)	(2)	(3)	(4)	(5)
Promoted	-0.771 (0.256)	-0.96 (0.256)	-0.535 (0.256)	-0.726 (0.279)	-0.642 (0.281)
Potential rating					
2 = Med	10.61 (0.294)	10.52 (0.292)	7.343 (0.282)	6.838 (0.299)	6.859 (0.299)
3 = High	20.70 (0.865)	15.50 (0.864)	14.66 (0.831)	13.57 (0.850)	13.52 (0.851)
Performance rating					
2 = Med	6.856 (0.329)	6.328 (0.303)	5.521 (0.536)	5.878 (0.536)	
3 = High	10.09 (0.417)	10.71 (0.540)	10.38 (0.543)	10.30 (0.569)	
Fiscal year FEs	Yes	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes
Location FEs	Yes	Yes	Yes	Yes	Yes
Business unit FEs	Yes	Yes	Yes	Yes	Yes
Observations	900,209	900,209	900,209	900,209	900,209

Notes: This table is identical to Table 2, except for the addition of control variables for the worker's potential rating. A comparison of the coefficient on the female indicator with the same coefficient in Table 2 offers an estimate of the portion of the gender promotion gap that can be explained by potential ratings. Standard errors are clustered by worker.

Read:

- Figure 3 shows women are more likely to receive high performance ratings but less likely to receive medium or high potential ratings.
- Table 4 confirms this in regressions: with controls, women receive higher performance ratings but lower potential ratings.
- Table 5 shows that adding potential ratings to the promotion regression explains about half of the overall gender promotion gap.
- A one-point increase in potential predicts a larger jump in promotion probability than a one-point increase in performance.

Economic message. The firm's promotion technology is heavily tilted toward who is seen as having upward-looking leadership potential, not just who is performing well today.

5.4 Table 6: realized potential versus assessed potential

Read:

- High current potential does predict higher future performance, so potential ratings contain some real information.
- But conditional on current potential and performance, women receive higher future performance ratings than comparable men.

- The same is true among promoted workers, so the result is not just about doing well in the old job.
- Yet women still receive lower future potential ratings than men with the same initial ratings.

Economic message. This is the core evidence of bias: women outperform the firm's forecast, but the firm does not revise its forecast upward.

	Full sample		Promoted sample	
Next performance rating	(1)	(2)	(3)	(4)
<i>Panel A</i>				
Female	0.0328 (0.0046)	0.0196 (0.0049)	0.0285 (0.0154)	0.0279 (0.0154)
Potential rating				
2 = Med	0.0913 (0.0048)	0.1025 (0.0052)	0.0678 (0.0162)	0.0665 (0.0163)
3 = High	0.1677 (0.0116)	0.197 (0.0118)	0.1266 (0.0278)	0.1275 (0.0282)
Performance rating				
2 = Med	0.3637 (0.0111)	0.2608 (0.0116)	0.2697 (0.0522)	0.2609 (0.0513)
3 = High	0.7671 (0.0121)	0.5776 (0.0126)	0.5139 (0.0534)	0.4975 (0.0524)
Fiscal year FEs	Yes	Yes	Yes	Yes
Demographic controls		Yes		Yes
Location FEs		Yes		
Business unit FEs		Yes		
Observations	586,338	586,338	5,222	5,222
Next potential rating	(5)	(6)	(7)	(8)
<i>Panel B</i>				
Female	-0.0482 (0.0047)	-0.0305 (0.005)	-0.0685 (0.018)	-0.0536 (0.0175)
Potential rating				
2 = Med	0.4241 (0.0055)	0.2858 (0.0059)	0.2461 (0.0186)	0.2074 (0.0184)
3 = High	0.7297 (0.0167)	0.5377 (0.0164)	0.4593 (0.0359)	0.3816 (0.0353)
Performance rating				
2 = Med	0.2135 (0.0091)	0.1659 (0.0096)	0.1082 (0.0592)	0.0775 (0.062)
3 = High	0.3147 (0.0099)	0.2922 (0.0106)	0.1795 (0.0602)	0.175 (0.063)
Fiscal year FEs	Yes	Yes	Yes	Yes
Demographic controls		Yes		Yes
Location FEs		Yes		
Business unit FEs		Yes		
Observations	586,338	586,338	5,222	5,222

Notes: Panel A reports a regression of Nine Box performance ratings 12 months in the future on control variables as described in Table 2. Columns 1 and 2 include the 19,640 workers with observable future ratings, which excludes workers who leave or are hired in the last evaluation year. Columns 3 and 4 restrict the sample to 5,222 worker-months corresponding to promotion events and do not control for location fixed effects due to the smaller sample size; this includes 4,344 workers, some of whom receive more than one promotion. Panel B is identical to panel A except that the dependent variable is the Nine Box potential rating 12 months in the future. Standard errors are clustered by worker.

	Full sample			Age ≤ 35		
	Leave	Potential	Potential	Leave	Potential	Potential
	(1)	(2)	(3)	(4)	(5)	(6)
Female	5.431 (0.430)	-0.0856 (0.00562)	-0.0895 (0.00582)	12.52 (0.906)	-0.0449 (0.0108)	-0.0382 (0.0113)
Past leaves			-0.0392 (0.00630)			-0.0339 (0.0127)
Female × Past leaves			0.0363 (0.00821)			0.00291 (0.0160)
Future leaves			-0.0422 (0.00449)			-0.0334 (0.0143)
Female × Future leaves			0.0300 (0.00607)			0.0180 (0.0162)
Fiscal year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Performance controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	886,899	886,899	886,899	214,040	214,040	214,040
DV mean	8.203	1.431	1.431	8.864	1.631	1.631

Notes: Column 1 presents a regression of whether a worker takes a leave of absence in the next month on the female indicator and other control variables as described in Table 2. Leaves take values of 0 or 1,200 so that coefficients can be interpreted as annual percents. Columns 2 and 3 relate gender and leaves to Nine Box potential ratings. Past and future leaves measure the number of months of leave a worker has taken in their past or future history with the firm within our data sample. Columns 4–6 repeat the exercise for workers aged 35 and under. Standard errors are clustered by worker.

5.5 Table 7: leaves of absence

Read:

- Women, especially younger women, are more likely to take leave.
- Past and future leave are associated with lower potential ratings.
- However, the leave penalty is less severe for women than for men, so controlling for leave does not explain much of the overall female potential gap.
- Among workers age 35 and under, leave explains only a modest portion of the gap.

Economic message. Career interruptions are not the main reason women are rated lower on potential in this firm.

5.6 Table 8 and Figure 4: attrition, risk of loss, and strategic retention

TABLE 8—ATTRITION AND RISK OF LOSS

	Attrition (1)	Risk of loss (1–3) (2)	Next potential (3)	Promotion (4)	Next log salary (5)	Next performance (6)
<i>Panel A. All ages</i>						
Female	–2.2253 (0.4267)	–0.0547 (0.0068)	–0.0432 (0.0056)	–0.9293 (0.3144)	–0.1422 (0.0069)	0.0319 (0.0056)
Risk of loss						
2 = Med			0.0834 (0.0064)	0.5869 (0.3587)	0.0624 (0.008)	0.0095 (0.0061)
3 = High			0.0918 (0.0136)	3.8923 (0.7575)	–0.007 (0.016)	–0.007 (0.0127)
Fiscal year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Ratings FEs	Yes	Yes	Yes	Yes	Yes	Yes
Workers	24,122	24,122	18,120	24,122	11,585	18,120
Observations	533,780	533,780	415,683	533,780	135,850	415,683
DV mean	22.657	1.421	1.382	10.852	11.052	2.189
	(7)	(8)	(9)	(10)	(11)	(12)
<i>Panel B. Age ≤ 35</i>						
Female	0.176 (1.0174)	–0.0105 (0.0149)	–0.0429 (0.0145)	–3.2353 (0.8477)	–0.0036 (0.0136)	0.0444 (0.0123)
Risk of loss						
2 = Med			0.0732 (0.0153)	–1.6387 (0.8983)	0.087 (0.0143)	0.0081 (0.0129)
3 = High			0.0588 (0.0287)	2.805 (1.6858)	0.1108 (0.027)	0 (0.0235)
Fiscal year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Ratings FEs	Yes	Yes	Yes	Yes	Yes	Yes
Workers	7,086	7,086	4,804	7,086	2,398	4,804
Observations	120,003	120,003	84,951	120,003	26,441	84,951
DV mean	29.869	1.513	1.623	18.93	10.925	2.19

Notes: The dependent variables for attrition and promotion take a value of 1,200 if the worker leaves or is promoted in the following month, and 0 otherwise so that coefficients represent annualized percents. Column 1 reports regressions of actual attrition in the next month on the female indicator. Column 2 adds controls for “risk of loss” ratings assigned by the firm (the omitted category is 1, low risk of loss). Columns 3–6 examine the relationship between risk of loss, gender, 12-month-ahead potential rating, whether a worker is promoted in the following month, and 12-month-ahead salary and performance ratings. The sample omits the last month that a worker’s location is in our sample to allow for observations of future behavior. Panel B restricts the sample to workers 35 and under. Standard errors are clustered by worker.

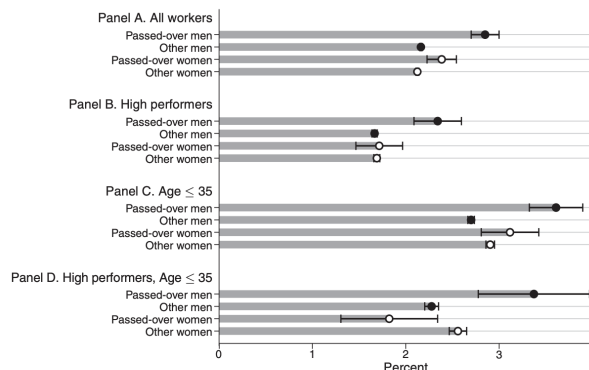


FIGURE 4. ATTRITION AMONG WORKERS PASSED OVER FOR PROMOTION

Notes: The figure presents predictive margins for the probability of attrition in the next month from four linear probability models (one for each panel), for four groups: passed-over men, other men, passed-over women, and other women. A worker is considered to have been “passed over” if another worker sharing the same manager is promoted in the next month but the focal worker is not. All regressions control for whether a promotion event is within the team, worker Nine Box ratings, and fiscal year fixed effects. Brackets report standard errors. Panel A includes all 29,809 workers in the full sample. Panel B restricts the sample to 10,831 workers who received a Nine Box performance rating of 3 (high performance). Panel C restricts the sample to 10,033 workers who were age 35 or under. Panel D restricts to 3,062 who were both 35 or under as well as high performers.

Read:

- Women are less likely to leave the firm than men, and managers appear to know this because women receive lower risk-of-loss ratings.
- Workers assessed as higher risk of loss receive higher next-period potential ratings, higher promotion rates, and higher future pay.
- Those same workers do *not* deliver higher future performance ratings.
- Figure 4 shows that men are much more likely than women to leave after being passed over for promotion. The difference is especially large among high performers and among workers under 35.

Economic message. The firm seems to use potential ratings partly as a retention instrument. Since threatened exit is concentrated more among men, this channel mechanically advantages men.

5.7 Tables 9–10: desirable promotions and local gender norms

Read:

- Table 9 shows gender gaps persist even for promotions that stay in the same metropolitan area, involve large raises, or do not add subordinates.
- Potential ratings still explain a substantial share of these gaps.

Promotion	Full sample		Same MSA		Raise > 25		No subordinates	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female	−1.837 (0.266)	−0.963 (0.256)	−0.612 (0.237)	0.098 (0.230)	−0.348 (0.128)	−0.198 (0.128)	−2.491 (0.387)	−1.476 (0.372)
Performance rating								
2 = Med	6.498 (0.325)	6.856 (0.329)	5.204 (0.285)	5.497 (0.289)	1.681 (0.160)	1.656 (0.160)	6.701 (0.504)	8.091 (0.521)
3 = High	11.345 (0.424)	10.092 (0.417)	9.059 (0.372)	8.046 (0.365)	1.795 (0.195)	1.473 (0.192)	10.740 (0.641)	10.771 (0.641)
Potential rating								
2 = Med		10.521 (0.292)		8.550 (0.260)		1.455 (0.137)		11.362 (0.437)
3 = High		19.498 (0.864)		15.780 (0.783)		4.497 (0.444)		20.671 (1.414)
Fiscal year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	900,209	900,209	900,209	900,209	900,209	900,209	426,989	426,989
DV mean	11.949	11.949	9.447	9.447	2.954	2.954	12.023	12.023

Notes: These regressions examine the relation between desirable promotions, gender, and performance and potential ratings. The dependent variables are indicators for promotions that allow the worker to remain in the same metropolitan statistical area (MSA) after promotion (or state if not within an MSA), promotions that involve a greater than 25 percent raise in salary, and promotions that do not involve the management of subordinates. The sample in columns 7 and 8 is limited to workers currently managing zero subordinates (so promotions involve no change in the number of subordinates). Standard errors are clustered by worker.

	DV: Potential rating			DV: Performance rating		
	(1)	(2)	(3)	(4)	(5)	(6)
Female	0.122 (0.083)	0.059 (0.052)	−0.271 (0.047)	−0.048 (0.084)	−0.029 (0.052)	−0.116 (0.049)
County management gap				−0.172 (0.091)		
Female × County management gap				−0.355 (0.141)	0.141 (0.143)	
County pay gap		−0.349 (0.026)			−0.074 (0.025)	
Female × County pay gap		−0.103 (0.037)			0.047 (0.037)	
County female edu attainment			0.525 (0.052)			0.101 (0.050)
Female × County female edu attainment			0.292 (0.074)			0.234 (0.077)
Fiscal year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	897,439	897,439	897,439	897,439	897,439	897,439

Notes: This table shows how the gender gap in potential and performance ratings varies with county-level characteristics, measured following the methodology in the United Nations Human Development Reports (2021). *County management gap* is the fraction of men among workers with management standard occupational classification (SOC) codes. *County pay gap* is men’s median earnings divided by women’s median earnings. *County female educational attainment* is the fraction of women over the age of 18 with at least some college education. County data come from the American Community Survey (Ruggles et al. 2025). Standard errors are clustered by worker.

- Table 10 shows the female potential gap is larger in counties with fewer women in management, larger gender wage gaps, and lower female educational attainment.
- Comparable interactions are weak for current performance ratings.

Economic message. The evidence is hard to reconcile with a simple “women want promotion less” story. It fits better with a story in which managers struggle to imagine women as future leaders in less gender-equal environments.

5.8 Figure 5 and Tables 11–12: misallocation, manager heterogeneity, and policy trade-offs

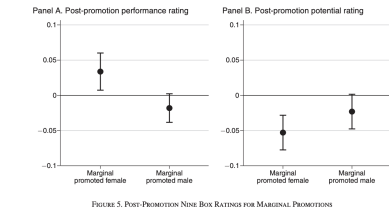


FIGURE 5. POST-PROMOTION NINE BOX RATINGS FOR MARGINAL PROMOTIONS

Notes: This figure reports estimates for the IV coefficient on the promoted indicator from equation (3), as described in the text. Panel A focuses on a promoted worker’s 12-month-ahead performance rating, whereas panel B focuses on that worker’s 12-month-ahead potential rating. Vertical brackets represent standard errors.

Potential rating	(1)	(2)	(3)	(4)
Female	−0.0566 (0.00583)	−0.0606 (0.0226)	−0.0211 (0.0152)	−0.0312 (0.0102)
Manager characteristic				
	−0.0288 (0.00725)	0.00172 (0.000337)	0.0314 (0.00460)	0.0331 (0.00397)
Female × Manager characteristic	0.0219 (0.00961)	0.00023 (0.000470)	−0.0132 (0.00664)	−0.0118 (0.00596)
This model’s manager characteristic		Manager age	Manager’s performance rating (1–3)	Manager’s potential rating (1–3)
Fiscal year FEs	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
Location FEs	Yes	Yes	Yes	Yes
Business unit FEs	Yes	Yes	Yes	Yes
Observations	897,520	897,520	829,429	829,429

Notes: Column 1 examines heterogeneity in the focal worker’s potential ratings by the manager’s gender. Column 2 examines variation by a linear term for age in years. Columns 3 and 4 examine variation by a linear term for the manager’s own performance and potential rating, respectively. Columns 1 and 2 include 29,787 workers and omit 2,707 worker-months for which the manager’s demographics are incomplete. Columns 3 and 4 include 29,510 workers and omit 70,801 worker-months when the manager had not yet received a Nine Box rating. Standard errors are clustered by worker.

	Promotion rates		Expected next performance rating among promoted	
	Among men (1)	Among women (2)	Full sample (3)	True promoted sample (4)
Baseline	12.6332 (0.1815)	10.9882 (0.1426)	2.2933 (0.0045)	2.2743 (0.0091)
current promotion policy				
Counterfactual	12.2006 (0.1492)	11.5865 (0.1426)	2.2772 (0.0041)	2.2633 (0.0092)

Notes: This table reports expected promotion rates and expected future performance ratings under the firm’s current promotion policy and a counterfactual promotion that ignores information on gender and potential. Details are provided in Section VIII. Columns 1 and 2 provide the expectations of promotion rates for men and women, respectively. Column 3 considers the full set of workers present in the next year regardless of whether they were actually promoted in the true sample and reports expectations of the 12-month-ahead performance ratings among the counterfactual set of promoted workers by weighting by the counterfactual’s estimated promotion probabilities. Column 4 does the same but restricts the weighted mean of 12-month-ahead performance to workers in the true promoted sample. When estimating counterfactual 12-month-ahead performance, the baseline policy uses predicted values of promotion rates based on gender, performance ratings, potential ratings, year, age, tenure, and race/ethnicity indicators. The counterfactual uses the baseline policy but omits gender and potential ratings when estimating promotion rates. Bootstrapped standard errors clustered by worker are in parentheses.

Read:

- Figure 5 shows marginally promoted women have higher future performance than marginally promoted men.
- The same women continue to receive lower future potential ratings even after promotion.
- Table 11 shows female managers reduce the gender potential gap somewhat, but they rate everyone lower on potential; older managers do not materially reduce the gap, and highly rated managers tend to have larger gaps.
- Table 12 shows that ignoring gender and potential in promotions would reduce the gender promotion gap by about 65%, but it would also lower the average future performance of promoted workers.

Economic message. Two things are true at once: the current system is biased against women, and potential ratings still contain useful information. Good policy therefore requires debiasing, not simply erasing all forward-looking evaluation.

6 Short summary

- The paper studies a large retail firm's Nine Box system, which separately rates current performance and future potential.
- Women receive higher performance ratings but lower potential ratings.
- Women are less likely to be promoted, and differences in potential ratings explain roughly half of that promotion gap.
- Lower female potential ratings are not justified by future outcomes: women later outperform comparable men.
- Managers nonetheless continue to rate women as having lower potential in the next period.
- Strategic retention helps explain part of the gap because men are more likely to leave when passed over, and managers reward high quit risk with better ratings and opportunities.
- Stereotyping also appears important because the female potential gap is larger in less gender-equal local environments.
- An IV outcomes test shows marginally promoted women outperform marginally promoted men, suggesting underpromotion of women at the margin.
- Ignoring potential ratings altogether would improve equity but reduce the average future quality of promoted workers.
- The central policy challenge is to keep the information content of potential ratings while removing their gender bias.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that subjective assessments of worker potential are an economically important source of gender gaps in promotion and pay. Women are not being held back because they perform worse. On the contrary, they perform better on average. The gap arises because firms systematically assess women as having less future upside, and those assessments strongly shape promotion decisions.

7.2 Economic intuition

Promotion decisions are about forecasts, not just records. That is exactly where bias can enter. Performance is anchored in observable output, but potential asks managers to imagine who looks like a future leader. If stereotypes, retention concerns, or organizational politics distort that imagination, then women can be underrated even when their realized performance is stronger.

7.3 Takeaway

The broad lesson is that firms should worry not only about biased evaluations of current work, but also about biased beliefs regarding future leadership. Because promotions, pay, and internal mobility often depend on those beliefs, distorted assessments of potential can create large and persistent career gaps. The paper therefore points toward a practical organizational agenda: make assessments of potential more structured, more evidence-based, and less vulnerable to stereotypes or strategic manipulation.